1. Study Title

Beyond Accuracy: A Rigorous Multi-Dimensional Evaluation Framework for Machine Learning Model Robustness, Interpretability, and Clinical Utility in Diabetes Prediction

2. Abstract

This research project aims to develop and rigorously evaluate machine learning models for Type 2 diabetes prediction. It addresses critical methodological gaps in existing literature, such as over-reliance on single datasets and a lack of focus on clinical utility. The core of this study is a multi-dimensional evaluation framework that includes a three-tier dataset strategy (NHANES, BRFSS, PIMA) for robust validation, a comprehensive assessment of clinical utility using Decision Curve Analysis (DCA), and a novel analysis of the consistency between SHAP and LIME explainability methods. The ultimate goal is to provide a transparent, reproducible, and clinically actionable framework for developing trustworthy prediction models in healthcare.

3. Background and Rationale

The prediction of Type 2 diabetes is a critical area of public health research, with machine learning offering significant promise. However, the current body of literature is hampered by several methodological limitations that question the real-world validity and utility of published models. These limitations include:

- Over-reliance on the PIMA Indians Diabetes Dataset, which is small and not representative of the general population.
- Lack of rigorous external validation, leading to overly optimistic performance estimates that do not generalize.
- Insufficient assessment of clinical utility, failing to answer whether a model's predictions would lead to better clinical decisions.
- Superficial use of explainable AI (XAI), where explanations are presented without assessing their reliability or consistency.

This study directly addresses these gaps by proposing and executing a comprehensive framework that prioritizes robustness, clinical utility, and trustworthy explainability.

4. Research Aims & Hypotheses

Aims:

- To develop and internally validate four machine learning models (Logistic Regression, Random Forest, XGBoost, SVM) for diabetes prediction using a large, nationally representative dataset (NHANES) .

- To assess the generalizability and robustness of these models through rigorous external validation on a separate, large-scale dataset (BRFSS).
- To evaluate and compare the clinical utility of the models using calibration analysis and Decision Curve Analysis (DCA).
- To conduct a novel consistency analysis of two prominent XAI methods (SHAP and LIME) to assess the reliability of their explanations.

Hypotheses:

- **H1: Predictive Performance.** The XGBoost model will achieve a statistically significant higher Area Under the ROC Curve (AUC) compared to the baseline Logistic Regression model when evaluated on the external validation dataset (BRFSS).
- **H2: Clinical Utility.** The best-performing prediction model will demonstrate a superior net benefit compared to the default strategies of "treat-all" and "treat-none" across a clinically relevant range of risk thresholds (5% to 50%), as measured by Decision Curve Analysis (DCA).
- **H3: Explainability Consistency.** For the tree-based models (Random Forest and XGBoost), the global feature importance rankings generated by SHAP and LIME will exhibit a strong, positive correlation, defined as a Spearman's rank correlation coefficient (ρ) > 0.7.

5. Methods

5.1. Study Design This study employs a

prospective model development and external validation design. It is a comparative computational study where four distinct algorithms are evaluated against each other on the same datasets.

5.2. Dataset Strategy

- **Primary Development Dataset: NHANES (2015-2020)**. Used for model training and internal validation. The outcome of diabetes will be defined by laboratory criteria (HbA1c ≥6.5% OR FPG ≥126 mg/dL) or physician diagnosis.
- **External Validation Dataset: BRFSS (2020-2022)**. Used for a single, one-time evaluation of the final trained models to assess generalizability. The outcome will be defined by self-reported physician diagnosis.
- **Reproducibility Baseline: PIMA Indians Dataset**. Used solely for benchmarking against published literature.

5.3. Preprocessing Pipeline A

scikit-learn Pipeline will be used to ensure consistency and prevent data leakage. Steps include:

- **Feature Mapping:** A detailed table will document the harmonization of variables across the three datasets.
- **Missing Data:** Missing continuous values will be imputed using the median; categorical values will be imputed using the mode.
- **Feature Scaling:** StandardScaler will be applied to all numerical features.
- **Class Imbalance:** Will be handled using class weights or threshold adjustment to avoid distorting probability distributions, which is a risk with oversampling.

5.4. Validation and Statistical Analysis

- **Internal Validation:** A **nested cross-validation** protocol will be used on the NHANES data (5-fold outer loop for performance estimation, 3-fold inner loop for hyperparameter tuning).
- **External Validation:** The best model of each type will be retrained on the full NHANES dataset and tested once on the untouched BRFSS dataset.
- **Statistical Comparison:** The **DeLong test** will be used to compare the AUCs of the models on the external validation predictions.

5.5. Assessment of Clinical Utility and Interpretability

- **Calibration:** Calibration plots and the Brier score will be used to assess the reliability of predicted probabilities.
- **Decision Curve Analysis (DCA):** DCA will be performed to quantify the net benefit of using the models across a range of clinical thresholds.
- **XAI Consistency:** The agreement between global feature importance rankings from SHAP and LIME will be quantified using Spearman's rank correlation.

6. Transparency and Reproducibility

This study is committed to open science principles. This protocol is pre-registered on the Open Science Framework (OSF). All analysis code, the feature mapping table, and an environment file will be made publicly available in a GitHub repository, which will be permanently archived on Zenodo. The final manuscript will adhere to the

TRIPOD-AI reporting guidelines, and a **PROBAST** assessment will be included to evaluate the risk of bias